

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 03 · Circles

With Answers

Questions with full worked solutions

- 46 pages
- Past papers up to January 2026

PURE 2 | TOPIC 03

Circles

Your classified route

19

QUESTIONS

154

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 03 | FAMILY 01

Circle Equation

WHAT THIS FAMILY ASKS

Recognise the circle equation demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

3 classified questions

26 total marks

T03

03_CENTRE

03_CENTRE_CORE

CHAPTER
Circles

FAMILY
Circle equation

VARIANT
core family move

12 marks

WMA12/01 OCTOBER 2022

Question 9

Past-paper question

WMA12-01_2022_Oct_Q09

9. In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

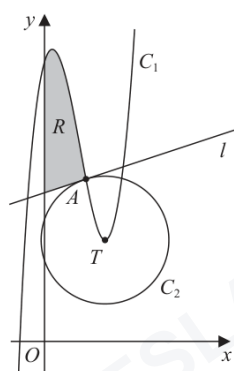


Figure 3

Figure 3 shows

- the curve C_1 with equation $y = x^3 - 5x^2 + 3x + 14$
- the circle C_2 with centre T

The point T is the minimum turning point of C_1

Using Figure 3 and calculus,

- (a) find the coordinates of T

(3)

The curve C_1 intersects the circle C_2 at the point A with x coordinate 2

- (b) Find an equation of the circle C_2

(3)

The line l shown in Figure 3, is the tangent to circle C_2 at A

- (c) Show that an equation of l is

$$y = \frac{1}{3}x + \frac{22}{3}$$

(3)

The region R , shown shaded in Figure 3, is bounded by C_1 , l and the y -axis.

- (d) Find the exact area of R .

(3)

T03 Circles

03_CENTRE Circle equation

03_CENTRE_CORE core
family move

WORKED SOLUTION

Question 9

WMA12/01 October 2022

WMA12-01_2022_Oct_Q09

1

Differentiate the cubicFor $y = x^3 - 5x^2 + 3x + 14$, $\frac{dy}{dx} = 3x^2 - 10x + 3$.

2

Find stationary x values $3x^2 - 10x + 3 = 0$ factors as $(3x - 1)(x - 3) = 0$, so $x = \frac{1}{3}$ or $x = 3$.

3

Identify the minimum $\frac{d^2y}{dx^2} = 6x - 10$. At $x = 3$, this is positive, so the minimum turning point has $x = 3$.

4

Find TSubstitute $x = 3$: $y = 27 - 45 + 9 + 14 = 5$. Hence $T = (3, 5)$.

5

Find AAt $x = 2$ on C_1 , $y = 8 - 20 + 6 + 14 = 8$, so $A = (2, 8)$.

6

Find the circle radiusThe centre of C_2 is $T = (3, 5)$ and it passes through $A = (2, 8)$. Thus $r^2 = (2 - 3)^2 + (8 - 5)^2 = 10$.

7

Write the circle equationTherefore C_2 has equation $(x - 3)^2 + (y - 5)^2 = 10$.

8

Find the tangent gradientThe radius TA has gradient $\frac{8-5}{2-3} = -3$, so the tangent at A has gradient $\frac{1}{3}$.

9

Write the tangent equationUsing $A = (2, 8)$, $y - 8 = \frac{1}{3}(x - 2)$, so $y = \frac{1}{3}x + \frac{22}{3}$.

10

Set up the areaFrom $x = 0$ to $x = 2$, the cubic is above the line. The required area is $\int_0^2 \left[x^3 - 5x^2 + 3x + 14 - \left(\frac{1}{3}x + \frac{22}{3} \right) \right] dx$.

11

Simplify the integrand

The integrand is $x^3 - 5x^2 + \frac{8}{3}x + \frac{20}{3}$.

12

Integrate

$$\left[\frac{x^4}{4} - \frac{5x^3}{3} + \frac{4x^2}{3} + \frac{20x}{3} \right]_0^2 = 4 - \frac{40}{3} + \frac{16}{3} + \frac{40}{3} = \frac{28}{3}.$$

FINAL ANSWER

WMA12-01_2022_Oct_Q09

(a) $T = (3, 5)$. (b) $(x - 3)^2 + (y - 5)^2 = 10$. (c) $y = \frac{1}{3}x + \frac{22}{3}$. (d) $\frac{28}{3}$.

DR ESLAM AHMED

T03

03_CENTRE

03_CENTRE_CORE

CHAPTER
Circles

FAMILY
Circle equation

VARIANT
core family move

6 marks

WMA12/01 JANUARY 2024

Question 3

Past-paper question

WMA12-01_2024_Jan_Q03

3. The circle C

- has centre $A(3, 5)$
- passes through the point $B(8, -7)$

(a) Find an equation for C .

(3)

The points M and N lie on C such that MN is a chord of C .

Given that MN

- lies above the x -axis
- is parallel to the x -axis
- has length $4\sqrt{22}$

(b) find an equation for the line passing through points M and N .

(3)

T03 Circles

03_CENTRE Circle equation

03_CENTRE_CORE core
family move

WORKED SOLUTION

Question 3

WMA12/01 January 2024

WMA12-01_2024_Jan_Q03

1

Find the radius squared

The centre is $A(3, 5)$ and the circle passes through $B(8, -7)$, so $r^2 = (8 - 3)^2 + (-7 - 5)^2 = 25 + 144 = 169$.

2

Write the circle equation

Hence C has equation $(x - 3)^2 + (y - 5)^2 = 169$.

3

Use the horizontal chord

Since MN is parallel to the x -axis, its equation has the form $y = c$.

4

Use half the chord length

The chord length is $4\sqrt{22}$, so the half-length is $2\sqrt{22}$.

5

Relate chord length to the circle

At height $y = c$, the horizontal half-length satisfies $169 - (c - 5)^2 = (2\sqrt{22})^2 = 88$.

6

Solve for c

$(c - 5)^2 = 81$, so $c = 14$ or $c = -4$.

7

Use the position above the x -axis

The chord lies above the x -axis, so $c = 14$. Therefore the line through M and N is $y = 14$.

FINAL ANSWER

WMA12-01_2024_Jan_Q03

(a) $(x - 3)^2 + (y - 5)^2 = 169$. (b) $y = 14$.

T03

03_CENTRE

03_CENTRE_CORE

CHAPTER
Circles

FAMILY
Circle equation

VARIANT
core family move

8 marks

WMA12/01 JANUARY 2025

Question 6

Past-paper question

WMA12-01_2025_Jan_Q06

6.

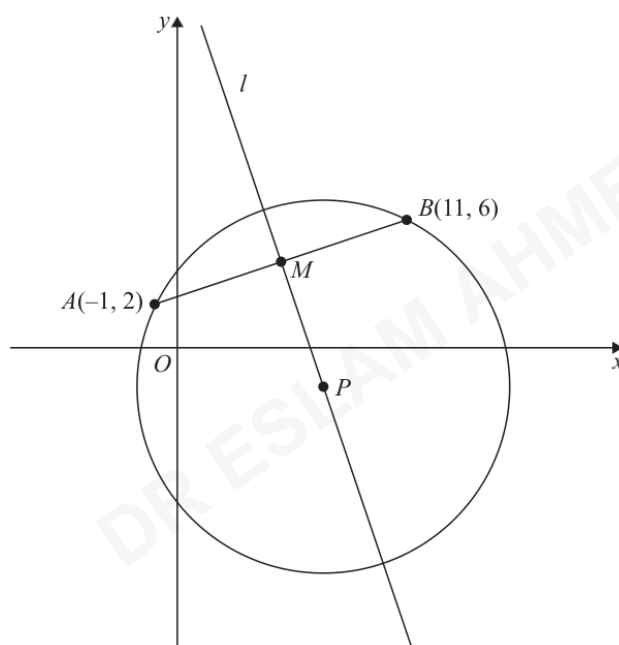


Figure 2

The point $A(-1, 2)$ and the point $B(11, 6)$ both lie on a circle with centre P .

The point M is the midpoint of AB .

Given that the line l passes through M and P , as shown in Figure 2,

- (a) find an equation for l , giving your answer in the form $y = mx + c$, where m and c are constants.

(4)

Given that P has coordinates $(7, k)$, where k is a constant,

- (b) find the value of k ,

(1)

- (c) find an equation for the circle.

(3)

T03 Circles

03_CENTRE Circle equation

03_CENTRE_CORE core
family move

WORKED SOLUTION

Question 6

WMA12/01 January 2025

WMA12-01_2025_Jan_Q06

1

Find the midpointFor $A(-1, 2)$ and $B(11, 6)$, the midpoint is $M\left(\frac{-1+11}{2}, \frac{2+6}{2}\right) = (5, 4)$.

2

Find the gradient of ABThe gradient of AB is $\frac{6-2}{11-(-1)} = \frac{4}{12} = \frac{1}{3}$.

3

Use the perpendicular bisectorThe centre lies on the perpendicular bisector of the chord AB , so line l has gradient -3 and passes through $M(5, 4)$.

4

Find line l $y - 4 = -3(x - 5)$, so $y = -3x + 19$.

5

Use the centre coordinatesSince $P = (7, k)$ lies on l , $k = -3(7) + 19 = -2$.

6

Find the radius squaredThe centre is $(7, -2)$. Using point A , $r^2 = (7 - (-1))^2 + (-2 - 2)^2 = 8^2 + (-4)^2 = 80$.

7

Write the circle equationTherefore the circle is $(x - 7)^2 + (y + 2)^2 = 80$.**FINAL ANSWER**

WMA12-01_2025_Jan_Q06

(a) $y = -3x + 19$. (b) $k = -2$. (c) $(x - 7)^2 + (y + 2)^2 = 80$.

TOPIC 03 | FAMILY 02

Line-Circle Intersections

WHAT THIS FAMILY ASKS

Recognise the line-circle intersections demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 Tangency means the quadratic has one repeated root.

1 classified questions

12 total marks

T03

03_MEET

03_MEET_HV01

CHAPTER
CirclesFAMILY
Line-circle intersectionsVARIANT
Tangency means the
quadratic has one repeated
root.
12 marksWMA12/01 MAY/JUNE 2022
Question 10

Past-paper question

WMA12-01_2022_MayJune_Q10

10. The circle C has centre $X(3, 5)$ and radius r

The line l has equation $y = 2x + k$, where k is a constant.

- (a) Show that l and C intersect when

$$5x^2 + (4k - 26)x + k^2 - 10k + 34 - r^2 = 0 \quad (3)$$

Given that l is a tangent to C ,

- (b) show that $5r^2 = (k + p)^2$, where p is a constant to be found. (3)

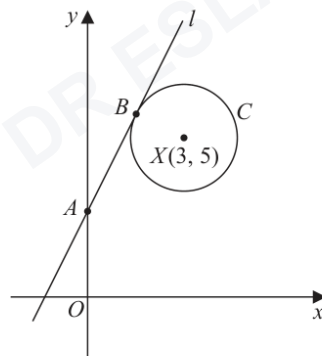


Figure 2

The line l

- cuts the y -axis at the point A
- touches the circle C at the point B

as shown in Figure 2.

Given that $AB = 2r$

- (c) find the value of k (6)

(Total 12 marks)

T03 Circles

03_MEET Line-circle intersections

03_MEET_HV01 Tangency means the quadratic has one repeated root.

WORKED SOLUTION

Question 10

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q10

1

Write the circle equationThe circle has centre $(3, 5)$ and radius r , so $(x - 3)^2 + (y - 5)^2 = r^2$.

2

Substitute the lineOn the line, $y = 2x + k$. Substitution gives $(x - 3)^2 + (2x + k - 5)^2 = r^2$.

3

Expand

$$x^2 - 6x + 9 + 4x^2 + 4kx - 20x + k^2 - 10k + 25 - r^2 = 0.$$

4

Show the intersection equationCollecting terms gives $5x^2 + (4k - 26)x + k^2 - 10k + 34 - r^2 = 0$.

5

Use tangency

For a tangent, the quadratic has one repeated root, so its discriminant is zero.

6

Set the discriminant to zero

$$(4k - 26)^2 - 20(k^2 - 10k + 34 - r^2) = 0.$$

7

SimplifyThis reduces to $20r^2 - 4k^2 - 8k - 4 = 0$, hence $5r^2 = (k + 1)^2$. Therefore $p = 1$.

8

Find the tangent point x coordinateFor the repeated root, $x_B = -\frac{4k-26}{2 \cdot 5} = \frac{13-2k}{5}$.

9

Use distance ABThe point A is $(0, k)$ and B lies on a line of gradient 2, so $AB = \sqrt{x_B^2 + (2x_B)^2} = \sqrt{5}x_B$.

10

Use the radius relationFrom $5r^2 = (k + 1)^2$, $r = \frac{k+1}{\sqrt{5}}$ in the shown configuration.

11

Apply AB equals two r

$$\sqrt{5} \left(\frac{13-2k}{5} \right) = 2 \left(\frac{k+1}{\sqrt{5}} \right).$$

12

Solve for k

Multiplying by $\sqrt{5}$ gives $13 - 2k = 2k + 2$, so $k = \frac{11}{4}$.

FINAL ANSWER

WMA12-01_2022_MayJune_Q10

(a) **Proven.** (b) $5r^2 = (k+1)^2$, so $p = 1$. (c) $k = \frac{11}{4}$.

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TOPIC 03 | FAMILY 03

Radius/Distance

WHAT THIS FAMILY ASKS

Recognise the radius/distance demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

7 classified questions

49 total marks

T03

03_RADIUS

03_RADIUS_CORE

CHAPTER
Circles

FAMILY
Radius/distance

VARIANT
core family move

8 marks

WMA12/01 JANUARY 2020

Question 6

Past-paper question

WMA12-01_2020_Jan_Q06

6. The circle C has equation

$$x^2 + y^2 + 6x - 4y - 14 = 0$$

- (a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact radius of C .

(3)

The line with equation $y = k$, where k is a constant, is a tangent to C .

- (b) Find the possible values of k .

(2)

The line with equation $y = p$, where p is a negative constant, is a chord of C .

Given that the length of this chord is 4 units,

- (c) find the value of p .

(3)

T03 Circles

03_RADIUS Radius/distance

03_RADIUS_CORE core
family move

WORKED SOLUTION

Question 6

WMA12/01 January 2020

WMA12-01_2020_Jan_Q06

1

Complete the square

$$x^2 + 6x = (x + 3)^2 - 9 \text{ and } y^2 - 4y = (y - 2)^2 - 4.$$

2

Find the centre and radius

The circle becomes $(x + 3)^2 + (y - 2)^2 = 27$. Hence the centre is $(-3, 2)$ and the radius is $\sqrt{27} = 3\sqrt{3}$.

3

Use horizontal tangents

A horizontal tangent has equation $y = k$. The possible tangent levels are one radius above and below the centre, so $k = 2 \pm 3\sqrt{3}$.

4

Use half the chord length

The chord has length 4, so half the chord is 2. If the chord is $y = p$, its perpendicular distance from the centre is $|p - 2|$.

5

Apply Pythagoras

$$(p - 2)^2 + 2^2 = (3\sqrt{3})^2 = 27, \text{ so } (p - 2)^2 = 23.$$

6

Choose the negative value

Since p is negative, $p = 2 - \sqrt{23}$.

FINAL ANSWER

WMA12-01_2020_Jan_Q06

(a)(i) $(-3, 2)$. (a)(ii) $3\sqrt{3}$. (b) $k = 2 \pm 3\sqrt{3}$. (c) $p = 2 - \sqrt{23}$.

T03

03_RADIUS

03_RADIUS_CORE

CHAPTER
Circles

FAMILY
Radius/distance

VARIANT
core family move

8 marks

WMA12/01 OCTOBER 2021

Question 6

Past-paper question

WMA12-01_2021_Oct_Q06

6. (i) The circle C_1 has equation

$$x^2 + y^2 + 10x - 12y = k \quad \text{where } k \text{ is a constant}$$

- (a) Find the coordinates of the centre of C_1 (2)

- (b) State the possible range in values for k . (2)

- (ii) The point $P(p, 0)$, the point $Q(-2, 10)$ and the point $R(8, -14)$ lie on a different circle, C_2

Given that

- p is a positive constant
- QR is a diameter of C_2

find the exact value of p .

(4)

T03 Circles

03_RADIUS Radius/distance

03_RADIUS_CORE core
family move

WORKED SOLUTION

Question 6

WMA12/01 October 2021

WMA12-01_2021_Oct_Q06

1

Complete the square

$$x^2 + 10x = (x + 5)^2 - 25 \text{ and } y^2 - 12y = (y - 6)^2 - 36.$$

2

Write centre-radius form

$$\text{The circle becomes } (x + 5)^2 + (y - 6)^2 = k + 61.$$

3

Read the centre

The centre is $(-5, 6)$.

4

State the range for k

For a circle with positive radius, $k + 61 > 0$, so $k > -61$.

5

Use the diameter property

If QR is a diameter and P lies on the circle, then $\angle QPR = 90^\circ$.

6

Use perpendicular vectors

So $(Q - P) \cdot (R - P) = 0$. With $P = (p, 0)$, $Q = (-2, 10)$, and $R = (8, -14)$, this gives $(-2 - p, 10) \cdot (8 - p, -14) = 0$.

7

Form the quadratic

$$(-2 - p)(8 - p) - 140 = 0, \text{ so } p^2 - 6p - 156 = 0.$$

8

Choose the positive value

$$p = \frac{6 \pm \sqrt{660}}{2} = 3 \pm \sqrt{165}. \text{ Since } p \text{ is positive, } p = 3 + \sqrt{165}.$$

FINAL ANSWER

WMA12-01_2021_Oct_Q06

(i)(a) $(-5, 6)$. (i)(b) $k > -61$. (ii) $p = 3 + \sqrt{165}$.

T03

03_RADIUS

03_RADIUS_CORE

CHAPTER
Circles

FAMILY
Radius/distance

VARIANT
core family move
6 marks

WMA12/01 MAY/JUNE 2024

Question 7

Past-paper question

WMA12-01_2024_MayJune_Q07

7. The circle C_1 has equation

$$x^2 + y^2 + 8x - 10y = 29$$

- (a) (i) Find the coordinates of the centre of C_1
(ii) Find the exact value of the radius of C_1

(3)

In part (b) you must show detailed reasoning.

The circle C_2 has equation

$$(x - 5)^2 + (y + 8)^2 = 52$$

- (b) Prove that the circles C_1 and C_2 neither touch nor intersect.

(3)

T03 Circles

03_RADIUS Radius/distance

03_RADIUS_CORE core
family move

WORKED SOLUTION

Question 7

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q07

1

Complete the square $x^2 + y^2 + 8x - 10y = 29$ becomes $(x + 4)^2 - 16 + (y - 5)^2 - 25 = 29$.

2

Write circle C1 in standard form $(x + 4)^2 + (y - 5)^2 = 70$.

3

Read centre and radiusThe centre of C_1 is $(-4, 5)$ and its radius is $\sqrt{70}$.

4

Read C2 C_2 has equation $(x - 5)^2 + (y + 8)^2 = 52$, so its centre is $(5, -8)$ and its radius is $\sqrt{52} = 2\sqrt{13}$.

5

Find distance between centresThe distance between $(-4, 5)$ and $(5, -8)$ is $\sqrt{9^2 + (-13)^2} = \sqrt{250} = 5\sqrt{10}$.

6

Compare with radii $\sqrt{70} + 2\sqrt{13} = 15.58 \dots$, while $5\sqrt{10} = 15.81 \dots$

7

Conclude

The distance between centres is greater than the sum of the radii, so the circles are separate. Therefore they neither touch nor intersect.

FINAL ANSWER

WMA12-01_2024_MayJune_Q07

(a)(i) $(-4, 5)$. (a)(ii) $\sqrt{70}$. (b) **The circles neither touch nor intersect.**

T03

03_RADIUS

03_RADIUS_CORE

CHAPTER
Circles

FAMILY
Radius/distance

VARIANT
core family move
5 marks

WMA12/01 MAY/JUNE (R) 2024

Question 3

Past-paper question

WMA12-01_2024_MayJuneR_Q03

3. A circle has equation

$$x^2 + y^2 + 8x - 14y - 79 = 0$$

(a) Find

- (i) the coordinates of the centre of the circle,
- (ii) the radius of the circle.

(3)

Given that P is the point on the circle that is nearest the origin O ,

(b) find the exact length of OP

(2)

T03 Circles

03_RADIUS Radius/distance

03_RADIUS_CORE core
family move

WORKED SOLUTION

Question 3

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q03

1

Complete the square $x^2 + y^2 + 8x - 14y - 79 = 0$ becomes $(x + 4)^2 - 16 + (y - 7)^2 - 49 - 79 = 0$.

2

Write standard form $(x + 4)^2 + (y - 7)^2 = 144$.

3

Read centre and radiusThe centre is $(-4, 7)$ and the radius is 12.

4

Find distance from centre to originThe distance from $(-4, 7)$ to $O(0, 0)$ is $\sqrt{(-4)^2 + 7^2} = \sqrt{65}$.

5

Use origin inside the circleSince $\sqrt{65} < 12$, the origin is inside the circle.

6

Find nearest distance to the circleThe nearest point on the circle lies along the line from the centre to the origin, so $OP = 12 - \sqrt{65}$.**FINAL ANSWER**

WMA12-01_2024_MayJuneR_Q03

 $(a)(i) (-4, 7).$ $(a)(ii) 12.$ $(b) 12 - \sqrt{65}.$

T03

03_RADIUS

03_RADIUS_CORE

CHAPTER
Circles

FAMILY
Radius/distance

VARIANT
core family move
7 marks

WMA12/01 MAY/JUNE 2025

Question 2

Past-paper question

WMA12-01_2025_MayJune_Q02

2: The line joining the points $(-2, 5)$ and $(4, 15)$ is the diameter of a circle C .

(a) Find an equation for C .

(5)

(b) Hence find the exact coordinates of the point on C that is nearest the x -axis.

(2)

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T03 Circles

03_RADIUS Radius/distance

03_RADIUS_CORE core
family move

WORKED SOLUTION

Question 2

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q02

1

Find the centre

The centre is the midpoint of $(-2, 5)$ and $(4, 15)$, so it is $\left(\frac{-2+4}{2}, \frac{5+15}{2}\right) = (1, 10)$.

2

Find the radius squared

Using $(-2, 5)$, $r^2 = (-2 - 1)^2 + (5 - 10)^2 = 9 + 25 = 34$.

3

Write the circle equation

Therefore C has equation $(x - 1)^2 + (y - 10)^2 = 34$.

4

Find the nearest point to the x -axis

The nearest point to the x -axis is vertically below the centre by one radius.

5

Write the coordinates

The radius is $\sqrt{34}$, so the point is $(1, 10 - \sqrt{34})$.

FINAL ANSWER

WMA12-01_2025_MayJune_Q02

(a) $(x - 1)^2 + (y - 10)^2 = 34$. (b) $(1, 10 - \sqrt{34})$.

T03

03_RADIUS

03_RADIUS_CORE

CHAPTER
CirclesFAMILY
Radius/distanceVARIANT
core family move

9 marks

WMA12/01 OCTOBER 2025

Question 6

Past-paper question

WMA12-01_2025_Oct_Q06

6.

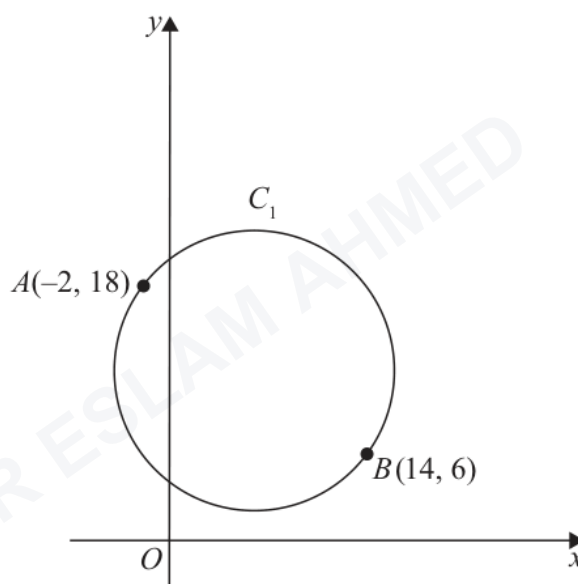


Figure 2

The points $A(-2, 18)$ and $B(14, 6)$ lie on a circle C_1 as shown in Figure 2.

Given that AB is a diameter of the circle C_1

(a) find an equation for C_1 making your method clear.

(5)

A circle C_2 has its centre at the origin.

Given that circles C_1 and C_2 touch,

(b) find possible equations for C_2

(4)

T03 Circles

03_RADIUS Radius/distance

03_RADIUS_CORE core
family move

WORKED SOLUTION

Question 6

WMA12/01 October 2025

WMA12-01_2025_Oct_Q06

1

Find the centre of C one

Since AB is a diameter, the centre is the midpoint of $A(-2, 18)$ and $B(14, 6)$. This is $(6, 12)$.

2

Find the radius squared

Using $A(-2, 18)$, $r^2 = (-2 - 6)^2 + (18 - 12)^2 = 64 + 36 = 100$.

3

Write C one

Therefore C_1 has equation $(x - 6)^2 + (y - 12)^2 = 100$.

4

Find the centre distance

The distance from the origin to $(6, 12)$ is $\sqrt{6^2 + 12^2} = 6\sqrt{5}$.

5

Find possible radii for C two

Since C_1 has radius 10, a circle centred at the origin can touch it with radius $6\sqrt{5} - 10$ or $6\sqrt{5} + 10$.

6

Write the equations

Thus C_2 can be $x^2 + y^2 = (6\sqrt{5} - 10)^2 = 280 - 120\sqrt{5}$ or $x^2 + y^2 = (6\sqrt{5} + 10)^2 = 280 + 120\sqrt{5}$.

FINAL ANSWER

WMA12-01_2025_Oct_Q06

(a) $(x - 6)^2 + (y - 12)^2 = 100$. (b) $x^2 + y^2 = 280 - 120\sqrt{5}$ or $x^2 + y^2 = 280 + 120\sqrt{5}$.

T03

03_RADIUS

03_RADIUS_CORE

CHAPTER
Circles

FAMILY
Radius/distance

VARIANT
core family move

6 marks

WMA12/01 JANUARY 2026

Question 5

Past-paper question

WMA12-01_2026_Jan_Q05

5. The circle C_1 has equation

$$x^2 + y^2 - 6x + 5y - 41 = 0$$

- (a) Find

- (i) the coordinates of the centre of C_1
- (ii) the radius of C_1

(3)

The circle C_2 has

- centre $(-k, 0)$ where k is a positive constant
- radius 5

Given that circles C_1 and C_2 touch

- (b) find the exact value of k .

(3)

T03 Circles

03_RADIUS Radius/distance

03_RADIUS_CORE core
family move

WORKED SOLUTION

Question 5

WMA12/01 January 2026

WMA12-01_2026_Jan_Q05

1

Complete the square

$$x^2 - 6x = (x - 3)^2 - 9 \text{ and } y^2 + 5y = \left(y + \frac{5}{2}\right)^2 - \frac{25}{4}.$$

2

Write the circle in centre-radius form

$$\text{The equation becomes } (x - 3)^2 + \left(y + \frac{5}{2}\right)^2 = \frac{225}{4}.$$

3

Read the centre and radius

$$\text{The centre is } \left(3, -\frac{5}{2}\right) \text{ and the radius is } \sqrt{\frac{225}{4}} = \frac{15}{2}.$$

4

Use the touching condition

The centre of C_2 is $(-k, 0)$ and its radius is 5. Since the circles touch externally, the distance between centres is $\frac{15}{2} + 5 = \frac{25}{2}$.

5

Solve for k

$$(k+3)^2 + \left(\frac{5}{2}\right)^2 = \left(\frac{25}{2}\right)^2, \text{ so } (k+3)^2 = 150. \text{ Since } k > 0, k+3 = 5\sqrt{6}, \text{ hence } k = 5\sqrt{6} - 3.$$

FINAL ANSWER

WMA12-01_2026_Jan_Q05

$$(a)(i) \left(3, -\frac{5}{2}\right). \quad (a)(ii) \frac{15}{2}. \quad (b) k = 5\sqrt{6} - 3.$$

TOPIC 03 | FAMILY 04

Tangents

WHAT THIS FAMILY ASKS

Recognise the tangents demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 If the radius is vertical, the tangent is horizontal.

8 classified questions

67 total marks

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core family move

7 marks

WMA12/01 OCTOBER 2019

Question 6

Past-paper question

WMA12-01_2019_Oct_Q06

6.

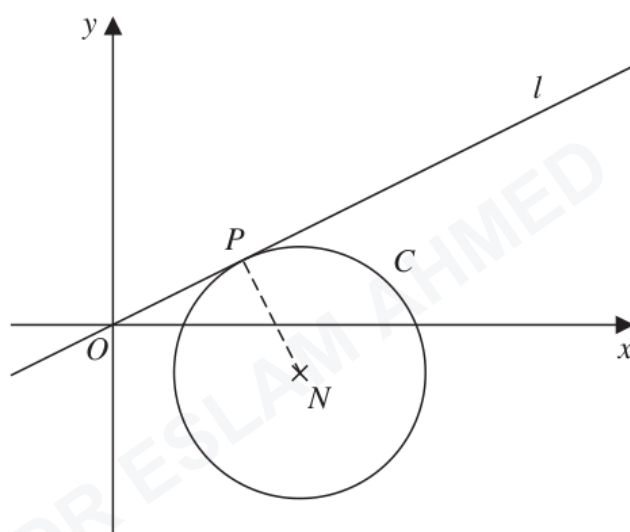


Figure 1

Figure 1 shows a sketch of a circle C with centre $N(4, -1)$.

The line l with equation $y = \frac{1}{2}x$ is a tangent to C at the point P .

Find

(a) the equation of line PN in the form $y = mx + c$, where m and c are constants, (2)

(b) the equation of C . (5)

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move

WORKED SOLUTION

Question 6

WMA12/01 October 2019

WMA12-01_2019_Oct_Q06

1

Use perpendicular gradients

The tangent has gradient $\frac{1}{2}$, so the radius PN has gradient -2 .

2

Find the equation of PN

Line PN passes through $N(4, -1)$, so $y + 1 = -2(x - 4)$. Therefore $y = -2x + 7$.

3

Find point P

At P , the tangent $y = \frac{1}{2}x$ meets the radius $y = -2x + 7$. Thus $\frac{1}{2}x = -2x + 7$, so $x = \frac{14}{5}$ and $y = \frac{7}{5}$.

4

Find the radius squared

$$r^2 = \left(\frac{14}{5} - 4\right)^2 + \left(\frac{7}{5} + 1\right)^2 = \left(-\frac{6}{5}\right)^2 + \left(\frac{12}{5}\right)^2 = \frac{36}{5}.$$

5

Write the circle equation

With centre $(4, -1)$ and radius squared $\frac{36}{5}$, the circle is $(x - 4)^2 + (y + 1)^2 = \frac{36}{5}$.

FINAL ANSWER

WMA12-01_2019_Oct_Q06

$$(a) y = -2x + 7. \quad (b) (x - 4)^2 + (y + 1)^2 = \frac{36}{5}.$$

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10 marks

WMA12/01 JANUARY 2021

Question 9

Past-paper question

WMA12-01_2021_Jan_Q09

9. A circle C has equation

$$(x - k)^2 + (y - 2k)^2 = k + 7$$

where k is a positive constant.

- (a) Write down, in terms of k ,

- (i) the coordinates of the centre of C ,
- (ii) the radius of C .

(2)

Given that the point $P(2, 3)$ lies on C

- (b) (i) show that $5k^2 - 17k + 6 = 0$

- (ii) hence find the possible values of k .

(3)

The tangent to the circle at P intersects the x -axis at point T .

Given that $k < 2$

- (c) calculate the exact area of triangle OPT .

(5)

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WORKED SOLUTION

Question 9

WMA12/01 January 2021

WMA12-01_2021_Jan_Q09

1

Read the centre and radius

From $(x - k)^2 + (y - 2k)^2 = k + 7$, the centre is $(k, 2k)$ and the radius is $\sqrt{k + 7}$.

2

Substitute point P

Since $P(2, 3)$ lies on the circle, $(2 - k)^2 + (3 - 2k)^2 = k + 7$.

3

Show the quadratic

Expanding gives $k^2 - 4k + 4 + 4k^2 - 12k + 9 = k + 7$, hence $5k^2 - 17k + 6 = 0$.

4

Solve for k

$5k^2 - 17k + 6 = (5k - 2)(k - 3)$, so $k = \frac{2}{5}$ or $k = 3$.

5

Use the condition k less than two

For the tangent calculation, $k < 2$, so the centre is $(\frac{2}{5}, \frac{4}{5})$.

6

Find the tangent gradient

The radius from the centre to P has gradient $\frac{3 - 4/5}{2 - 2/5} = \frac{11}{8}$, so the tangent gradient is $-\frac{8}{11}$.

7

Find the x intercept

The tangent is $y - 3 = -\frac{8}{11}(x - 2)$. Setting $y = 0$ gives $x = \frac{49}{8}$, so $T = (\frac{49}{8}, 0)$.

8

Calculate the area

Triangle OPT has base $\frac{49}{8}$ on the x -axis and height 3, so its area is $\frac{1}{2} \cdot \frac{49}{8} \cdot 3 = \frac{147}{16}$.

FINAL ANSWER

WMA12-01_2021_Jan_Q09

(a)(i) $(k, 2k)$. (a)(ii) $\sqrt{k + 7}$. (b)(ii) $k = 3$ or $\frac{2}{5}$. (c) $\frac{147}{16}$.

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8 marks

WMA12/01 JANUARY 2022

Question 6

Past-paper question

WMA12-01_2022_Jan_Q06

6.

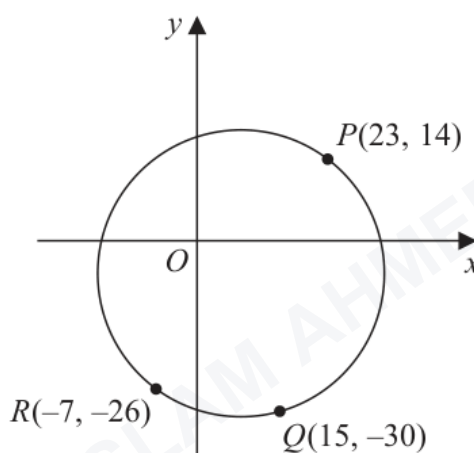


Figure 1

The points $P(23, 14)$, $Q(15, -30)$ and $R(-7, -26)$ lie on the circle C , as shown in Figure 1.

(a) Show that angle $PQR = 90^\circ$ (2)

(b) Hence, or otherwise, find

(i) the centre of C ,

(ii) the radius of C . (3)

Given that the point S lies on C such that the distance QS is greatest,

(c) find an equation of the tangent to C at S , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (3)

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WORKED SOLUTION

Question 6

WMA12/01 January 2022

WMA12-01_2022_Jan_Q06

1

Find the gradient QPThe gradient of QP is $\frac{14 - (-30)}{23 - 15} = \frac{44}{8} = \frac{11}{2}$.

2

Find the gradient QRThe gradient of QR is $\frac{-26 - (-30)}{-7 - 15} = \frac{4}{-22} = -\frac{2}{11}$.

3

Show the right angleThe product of the two gradients is -1 , so QP is perpendicular to QR and $\angle PQR = 90^\circ$.

4

Use the diameterSince the angle at Q is a right angle, PR is a diameter of the circle.

5

Find the centreThe centre is the midpoint of $P(23, 14)$ and $R(-7, -26)$, so it is $(8, -6)$.

6

Find the radius $PR = \sqrt{30^2 + 40^2} = 50$, so the radius is 25.

7

Find the farthest point from QThe greatest distance from Q occurs at the point opposite Q on the diameter through the centre. Thus $S = 2(8, -6) - (15, -30) = (1, 18)$.

8

Find the tangent gradientThe radius from the centre to S has gradient $\frac{18 - (-6)}{1 - 8} = -\frac{24}{7}$, so the tangent gradient is $\frac{7}{24}$.

9

Write the tangent equation $y - 18 = \frac{7}{24}(x - 1)$, which rearranges to $7x - 24y + 425 = 0$.**FINAL ANSWER**

WMA12-01_2022_Jan_Q06

(a) **Proven:** $\angle PQR = 90^\circ$. (b)(i) $(8, -6)$. (b)(ii) 25. (c) $7x - 24y + 425 = 0$.

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WMA12/01 JANUARY 2023

Question 6

Past-paper question

WMA12-01_2023_Jan_Q06

6. The circle C has equation

$$x^2 + y^2 + 8x - 4y = 0$$

- (a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact radius of C .

(3)

The point P lies on C .

Given that the tangent to C at P has equation $x + 2y + 10 = 0$

- (b) find the coordinates of P

(4)

- (c) Find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found.

(3)

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WORKED SOLUTION

Question 6

WMA12/01 January 2023

WMA12-01_2023_Jan_Q06

1

Complete the square $x^2 + y^2 + 8x - 4y = 0$ becomes $(x + 4)^2 - 16 + (y - 2)^2 - 4 = 0$.

2

Write the circle formSo $(x + 4)^2 + (y - 2)^2 = 20$.

3

Read the centre and radiusThe centre is $(-4, 2)$ and the radius is $\sqrt{20} = 2\sqrt{5}$.

4

Use the tangent gradientThe tangent $x + 2y + 10 = 0$ has gradient $-\frac{1}{2}$.

5

Find the radius gradient

The radius to the point of contact is perpendicular to the tangent, so its gradient is 2.

6

Write the normal through the centreThe radius line through the centre $(-4, 2)$ is $y - 2 = 2(x + 4)$, so $y = 2x + 10$.

7

Intersect with the tangentSubstitute $y = 2x + 10$ into $x + 2y + 10 = 0$: $x + 2(2x + 10) + 10 = 0$.

8

Find P $5x + 30 = 0$, so $x = -6$. Then $y = 2(-6) + 10 = -2$, hence $P = (-6, -2)$.

9

State the normalThe normal at P is the radius line already found, so its equation is $y = 2x + 10$.**FINAL ANSWER**

WMA12-01_2023_Jan_Q06

(a)(i) $(-4, 2)$. (a)(ii) $2\sqrt{5}$. (b) $P = (-6, -2)$. (c) $y = 2x + 10$.

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WMA12/01 MAY/JUNE 2023

Question 3

Past-paper question

WMA12-01_2023_MayJune_Q03

3. A circle C has centre $(2, 5)$

Given that the point $P(8, -3)$ lies on C

- (a) (i) find the radius of C

- (ii) find an equation for C

(3)

- (b) Find the equation of the tangent to C at P giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

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WORKED SOLUTION

Question 3

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q03

1

Use the distance formula

The centre is $(2, 5)$ and $P = (8, -3)$. The radius is $\sqrt{(8-2)^2 + (-3-5)^2}$.

2

Find the radius

$$r = \sqrt{6^2 + (-8)^2} = \sqrt{100} = 10.$$

3

Write the circle equation

With centre $(2, 5)$ and radius 10, the circle is $(x-2)^2 + (y-5)^2 = 100$.

4

Find the radius gradient

The gradient of the radius from the centre to P is $\frac{-3-5}{8-2} = \frac{-8}{6} = -\frac{4}{3}$.

5

Find the tangent gradient

The tangent is perpendicular to the radius, so its gradient is $\frac{3}{4}$.

6

Use point P

The tangent through $(8, -3)$ is $y + 3 = \frac{3}{4}(x - 8)$.

7

Put into integer form

Multiplying by 4 gives $4y + 12 = 3x - 24$, hence $3x - 4y - 36 = 0$.

FINAL ANSWER

WMA12-01_2023_MayJune_Q03

(a)(i) 10. (a)(ii) $(x-2)^2 + (y-5)^2 = 100$. (b) $3x - 4y - 36 = 0$.

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8 marks

WMA12/01 OCTOBER 2023

Question 7

Past-paper question

WMA12-01_2023_Oct_Q07

7.

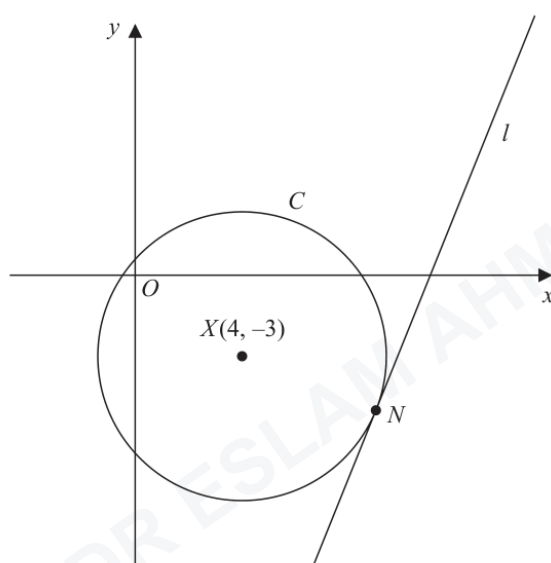


Figure 2

Figure 2 shows a sketch of

- the circle C with centre $X(4, -3)$
- the line l with equation $y = \frac{5}{2}x - \frac{55}{2}$

Given that l is the tangent to C at the point N ,(a) show that an equation for the straight line passing through X and N is

$$2x + 5y + 7 = 0$$

(3)

(b) Hence find

- the coordinates of N ,
- an equation for C .

(5)

T03 Circles

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WORKED SOLUTION

Question 7

WMA12/01 October 2023

WMA12-01_2023_Oct_Q07

1

Use perpendicular gradients

The tangent l has gradient $\frac{5}{2}$, so the radius XN is perpendicular to it and has gradient $-\frac{2}{5}$.

2

Use the centre

The line through $X(4, -3)$ with gradient $-\frac{2}{5}$ is $y + 3 = -\frac{2}{5}(x - 4)$.

3

Show the required line

Multiplying by 5 gives $5y + 15 = -2x + 8$, so $2x + 5y + 7 = 0$.

4

Intersect the radius and tangent

Use $y = \frac{5}{2}x - \frac{55}{2}$ in $2x + 5y + 7 = 0$.

5

Find x coordinate

$2x + 5\left(\frac{5}{2}x - \frac{55}{2}\right) + 7 = 0$. Multiplying by 2 gives $4x + 25x - 275 + 14 = 0$, so $29x = 261$ and $x = 9$.

6

Find y coordinate

$y = \frac{5}{2}(9) - \frac{55}{2} = \frac{45-55}{2} = -5$, so $N = (9, -5)$.

7

Find the radius squared

$XN^2 = (9 - 4)^2 + (-5 + 3)^2 = 5^2 + (-2)^2 = 29$.

8

Write the circle equation

With centre $(4, -3)$ and radius squared 29, the circle is $(x - 4)^2 + (y + 3)^2 = 29$.

FINAL ANSWER

WMA12-01_2023_Oct_Q07

(a) **Proven.** (b)(i) $N = (9, -5)$. (b)(ii) $(x - 4)^2 + (y + 3)^2 = 29$.

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10 marks

WMA12/01 OCTOBER 2024

Question 10

Past-paper question

WMA12-01_2024_Oct_Q10

10. The circle C has equation

$$x^2 + y^2 + 4x - 30y + 209 = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact value of the radius of C .

(3)

The line L has equation $y = mx + 1$, where m is a constant.

Given that L is the tangent to C at the point P ,

(b) show that

$$2m^2 - 7m - 22 = 0$$

(3)

(c) Hence find the possible pairs of coordinates of P .

(4)

T03 Circles

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WORKED SOLUTION

Question 10

WMA12/01 October 2024

WMA12-01_2024_Oct_Q10

1

Complete the square $x^2 + y^2 + 4x - 30y + 209 = 0$ becomes $(x + 2)^2 + (y - 15)^2 = 20$.

2

Read the centre and radiusThe centre is $(-2, 15)$ and the radius is $\sqrt{20} = 2\sqrt{5}$.

3

Use the tangent distanceThe line $y = mx + 1$ can be written as $mx - y + 1 = 0$. Its distance from $(-2, 15)$ must be $2\sqrt{5}$.

4

Set up the equation for m $\frac{|m(-2) - 15 + 1|}{\sqrt{m^2 + 1}} = 2\sqrt{5}$, so $\frac{|-2m - 14|}{\sqrt{m^2 + 1}} = 2\sqrt{5}$.

5

Show the quadraticSquaring gives $4(m+7)^2 = 20(m^2+1)$. Hence $(m+7)^2 = 5(m^2+1)$, so $2m^2 - 7m - 22 = 0$.

6

Find the possible gradients $2m^2 - 7m - 22 = 0$ gives $m = -2$ or $m = \frac{11}{2}$.

7

Use perpendicular radii

A tangent is perpendicular to the radius at the point of contact.

8

Find the first pointFor $m = -2$, the radius gradient is $\frac{1}{2}$. The radius through the centre is $y - 15 = \frac{1}{2}(x + 2)$. Solving with $y = -2x + 1$ gives $x = -6$ and $y = 13$.

9

Find the second pointFor $m = \frac{11}{2}$, the radius gradient is $-\frac{2}{11}$. The radius through the centre is $y - 15 = -\frac{2}{11}(x + 2)$. Solving with $y = \frac{11}{2}x + 1$ gives $x = \frac{12}{5}$ and $y = \frac{71}{5}$.

FINAL ANSWER

WMA12-01_2024_Oct_Q10

(a)(i) $(-2, 15)$. (a)(ii) $2\sqrt{5}$. (b) **Proven.** (c) $(-6, 13)$ and $\left(\frac{12}{5}, \frac{71}{5}\right)$.

DR ESLAM AHMED

T03

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If the radius is vertical, the
tangent is horizontal.

7 marks

WMA12/01 MAY/JUNE 2021

Question 6

Past-paper question

WMA12-01_2021_MayJune_Q06

6. A circle has equation

$$x^2 - 6x + y^2 + 8y + k = 0$$

where k is a positive constant.

Given that the x -axis is a tangent to this circle,

- (a) find the value of k .

(3)

The circle meets the coordinate axes at the points R , S and T .

- (b) Find the exact area of the triangle RST .

(4)

T03 Circles

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03_TANG_HV01 If the radius is vertical, the tangent is horizontal.

WORKED SOLUTION

Question 6

WMA12/01 May/June 2021

WMA12-01_2021_MayJune_Q06

1

Complete the square

$$x^2 - 6x = (x - 3)^2 - 9 \text{ and } y^2 + 8y = (y + 4)^2 - 16.$$

2

Write centre-radius form

$$\text{The circle becomes } (x - 3)^2 + (y + 4)^2 = 25 - k.$$

3

Use the tangent condition

The centre is $(3, -4)$. Since the x -axis is tangent, the radius is the distance from the centre to the x -axis, which is 4.

4

Find k

$$\text{Thus } 25 - k = 4^2 = 16, \text{ so } k = 9.$$

5

Find the intercepts

With $k = 9$, the circle is $(x - 3)^2 + (y + 4)^2 = 16$. On the x -axis, $y = 0$, giving $x = 3$.

6

Find the y -axis points

$$\text{On the } y\text{-axis, } x = 0, \text{ so } 9 + (y + 4)^2 = 16. \text{ Hence } y = -4 \pm \sqrt{7}.$$

7

Calculate the triangle area

The vertical base on the y -axis has length $2\sqrt{7}$, and its perpendicular distance to $(3, 0)$ is 3. The area is $\frac{1}{2}(2\sqrt{7})(3) = 3\sqrt{7}$.

FINAL ANSWER

$$(a) k = 9. \quad (b) 3\sqrt{7}.$$

WMA12-01_2021_MayJune_Q06